

Sheet 2: lin. alg.

1. (This formulation of law of demand seems to allow some goods' consumption to increase with price - i.e. Giffen goods - so long as that's offset by larger negative correlations elsewhere.)

Note that $\underline{x}(p) - \underline{x}(p') = \begin{pmatrix} \alpha_{1,m} \cdot \left(\frac{1}{p_1} - \frac{1}{p'_1} \right) \\ \vdots \\ \alpha_{n,m} \cdot \left(\frac{1}{p_n} - \frac{1}{p'_n} \right) \end{pmatrix}$

so $(\underline{x}(p) - \underline{x}(p')) \cdot (p - p') = \alpha_{1,m} \cdot \left(\frac{p_1}{p_1} - \frac{p_1}{p'_1} \right) + \dots + \alpha_{n,m} \cdot \left(\frac{p_n}{p_n} - \frac{p_n}{p'_n} \right)$

$$= -m \sum_{i=1}^n \alpha_i \cdot \left(\frac{p'_i}{p_i} + \frac{p_i}{p'_i} \right) < 0 \text{ in nontrivial cases (e.g. } m \neq 0)$$

$\underbrace{\alpha_i}_{\geq 0}$ as α_i weights non-neg.
 $\underbrace{\left(\frac{p'_i}{p_i} + \frac{p_i}{p'_i} \right)}_{> 0}$ for some i since prices $\in \mathbb{R}^+$ and distinct for some i .

2. a) $\left(\begin{array}{ccc|c} 1 & 3 & -2 & 2 \\ 1 & 2 & 1 & 1 \\ 1 & 5 & \alpha & \beta \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 3 & -2 & 2 \\ 0 & -1 & 3 & -1 \\ 0 & 2 & 2+\alpha & \beta-2 \end{array} \right)$ by Gaussian elimination.

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & -2 & 2 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & 8+\alpha & \beta-4 \end{array} \right)$$

From last row we have

$(8+\alpha)z = \beta-4.$

Also $y = 3z + 1$ and $x = -3y + 2z + 2$

- If $8+\alpha=0$ and $\beta-4 \neq 0 \Rightarrow \alpha=-8, \beta \neq 4$ then inconsistent. (no solns)

- If $\alpha=-8$ and $\beta=4$, infinite solutions.

$\hookrightarrow$ Specifically, $z=t, y=3t+1, x=-7t-1$

- Otherwise, unique solution $z = (\beta-4)/(8+\alpha), y = 3(\beta-4)/(8+\alpha) + 1, x = -7(\beta-4)/(8+\alpha) - 1.$

b) It depends on the values of α and β , in the same way as above. $b := \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$ is in the span of columns of $A := \begin{pmatrix} 1 & 3 & -2 \\ 1 & 2 & 1 \\ 5 & \alpha & \end{pmatrix}$ iff some linear combination of those columns can produce b - i.e., if the ^{system} equation $Ax = b$ has solutions.

So the answer is yes for $\alpha \neq -8$, and for $(\alpha = -9 \text{ AND } \beta = 4)$, and no otherwise. In the first case, these vectors form a basis for $\mathbb{R}^3$; in the second case there's linear dependence but b lies in the plane they span.

3a) W is the set of vectors w s.t. adding them to $\{u, v\}$ does not increase the ^{set's} span - i.e. all and only linear combinations of u, v .

So $W = \{ \lambda u + \mu v \}$ for $\lambda, \mu \in \mathbb{R}$; $= \left\{ \begin{pmatrix} \lambda + 2\mu \\ \lambda + \mu \\ \mu \end{pmatrix} \right\}$,
or all points in the plane $x - y - z = 0$.

b) As mentioned, we want to ensure $w \in W$ doesn't lie outside the span of $[u, v]$, otherwise it'd let us reach new vectors within span $[u, v, w]$. But this is precisely the vectors in the span we already have of $[u, v]$. So yes, they coincide.

c) Consider any point in $\mathbb{R}^3$, $r = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Let $z = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where $a - b - c \neq 0$, since $z \notin W$ and by part a).

Then we can write $r = \lambda u + \mu v + \gamma z = \begin{pmatrix} \lambda + 2\mu + \gamma a \\ \lambda + \mu + \gamma b \\ \mu + \gamma c \end{pmatrix}$

which, ^{can} solving we obtain solutions to, since the determinant $\det \begin{pmatrix} 1 & 2 & a \\ 1 & 1 & b \\ 0 & 1 & c \end{pmatrix} = -(b - a) + c(-1) = a - b - c \neq 0$ by assumption.

So u, v, z spans $\mathbb{R}^3$. And moreover, it's a minimal spanning set, since we require at least n vectors to span $\mathbb{R}^n$. So it's a basis.

This makes sense, since $\underline{u}, \underline{v}, \underline{z}$ are linearly independent vectors in $\mathbb{R}^3$, so together the three of them form a basis.

4. a)

~~A B B~~

$$A \cdot B = \begin{pmatrix} \sum_{i=1}^m a_{i1} b_{i1} & \dots & \sum_{i=1}^m a_{i1} b_{i\ell} \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^m a_{in} b_{i1} & \dots & \sum_{i=1}^m a_{in} b_{i\ell} \end{pmatrix}, \text{ which is } n \times \ell$$

and so $(A \cdot B)^T$ simply exchanges rows for columns, i.e. the first row will now ~~be~~ end with $\sum_{i=1}^m a_{ni} b_{i1}$, for an $\ell \times n$ matrix.

But this is exactly $B^T \cdot A^T = \begin{pmatrix} \sum_{i=1}^m b'_{i1} a'_{i1} & \dots & \sum_{i=1}^m b'_{i1} a'_{in} \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^m b'_{i\ell} a'_{i1} & \dots & \sum_{i=1}^m b'_{i\ell} a'_{in} \end{pmatrix}$ scalar mult. commutative

a $\ell \times n$ matrix

where now a', b' picks out entries in A^T, B^T , which 'just' exchange rows and columns in the original matrices, i.e. order of indexing.

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b) Recall $\det(A) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \cdot C_{ij}$ for some ~~col~~ j , with cofactors C .

But if we choose $j=1$, then we need only compute a single entry in the cofactor matrix, C_{11} .

$C'_{11} = M'_{11} = \det \begin{pmatrix} 2 & \dots \\ \vdots & 3 \end{pmatrix}$. Again, choose the first column and compute $2 \cdot C''_{11}$ for this n submatrix, ~~implying~~ equal to $3 \cdot C'''_{11}$.

We proceed until the final $\det \begin{pmatrix} n-1 & n-1 \\ 0 & n \end{pmatrix} = (n-1)n$, and then substitute backwards to find $C'_{11} = \prod_{i=1}^n n$.

So $\det(A) = n!$ i.e. product of all diagonal entries.

5a) Note that rank is at most 3.

$$\circ u_2 = u_1 + u_3.$$

$$\circ u_5 = 2u_3 - u_4$$

$$\circ u_3 = 2u_1 - u_4$$

So rank is 2; u_1 and u_4 are independent by inspection.

b) No, they don't span $\mathbb{R}^3$; we require at least 3 lin. indep vectors that but we only have 2.

The vectors span the subspace $\{ \mu u_1 + \lambda u_4 \}$, i.e. the points

$$r = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \mu + \lambda \\ \mu \\ \mu + \lambda \end{pmatrix} \Rightarrow \mu = y, \quad x = \lambda + y = z$$

So any ~~points~~ ^{vectors} not in the plane $x=z$ aren't in the span of $u_1, \dots, u_5$.

6c) Recall that $\det(A) \neq 0$ only if the rows ~~and~~ of A are linearly independent. But any 3×3 matrix constructed from u_i would be reduced-rank, i.e. have dependent rows. So their determinant is all zero: they map points in $\mathbb{R}^3$ to ~~the~~ a plane, scaling volumes by a factor zero.

6a) They'll both be true, since tr and $\det$ both true.

b) $\det(A) = 6$; $\text{tr}(A) = 5$.

$$\lambda_1'' \lambda_2$$

$$\lambda_1'' + \lambda_2$$

$$\text{So } \lambda_1 = 2, \lambda_2 = 3.$$

c) The eigenvalues λ solve $A \cdot x = \lambda x$, with char. polynomial $(A - \lambda I) \cdot x = 0$. We have nontrivial solutions to this homog.

$$\text{eq'n only if } \det(A - \lambda I) = 0 \Rightarrow (-1 - \lambda)(6 - \lambda) + 12 = 0.$$

$$\text{So } \lambda_1^2 - 5\lambda_1 + 6 = 0, \text{ which is solved by } \lambda = 2 \text{ or } \lambda = 3.$$

$$\therefore \begin{pmatrix} -1 & 2 \\ -6 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow -3x_1 + 2x_2 = 0, -6x_1 + 4x_2 = 0.$$

so $(2, 3)^T$ is an eigenvector

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$$\text{Let } S = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}; \quad S^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

$$\text{Then } S^{-1}A = \begin{pmatrix} 4 & -2 \\ -9 & 6 \end{pmatrix}; \quad S^{-1}AS = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = D \text{ as req'd.}$$

$$\text{ad) } A = SDS^{-1}. \quad \text{So } A^{10} = SD^{10}S^{-1} \text{ since } S^{-1} \cdot S = I \text{ cancel.}$$

$$\text{But } D^{10} = \begin{pmatrix} 2^{10} & 0 \\ 0 & 3^{10} \end{pmatrix}. \quad \text{So } A^{10} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2^{10} & 0 \\ 0 & 3^{10} \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

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better to prove this formally

3a). W is the set of vectors w s.t. adding them to $\{u, v\}$ does not increase the ^{set's} span - i.e. all and only linear combinations of u, v .

So $W = \{ \lambda u + \mu v \}$ for $\lambda, \mu \in \mathbb{R}$; $= \left\{ \begin{pmatrix} \lambda + 2\mu \\ \lambda + \mu \\ \mu \end{pmatrix} \right\}$,
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Q5 graded

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b) Recall $\det(A) = \sum_{j=1}^n a_{ij} \cdot C_{ij}$ for some ~~row~~ ^{col} j , with cofactors C .

But if we choose $j=1$, then we need only compute a single entry in the cofactor matrix, C_{11} .

$C'_{11} = M'_{11} = \det \begin{pmatrix} 2 & \dots \\ \vdots & 3 \end{pmatrix}$. Again, choose the first column and compute $2 \cdot C'_{11}$ for this n submatrix, ~~implying~~ equal to $3 \cdot C'_{11}$.

We proceed until the final $\det \begin{pmatrix} n-1 & n-1 \\ 0 & n \end{pmatrix} = (n-1)n$, and then substitute backwards to find $C'_{11} = \prod_{i=1}^n n$.

So $\det(A) = n!$ i.e. product of all diagonal entries.

5a) Note that rank is at most 3.

$$\circ u_2 = u_1 + u_3.$$

$$\circ u_5 = 2u_3 - u_4$$

Correct. Alternatively, simply note row 1 = row 3

$$\circ u_3 = 2u_1 - u_4$$

So rank is 2; u_1 and u_4 are independent by inspection.

b) No, they don't span $\mathbb{R}^3$; we require at least 3 lin. indep vectors that but we only have 2.

Make argument tighter by stating which vectors are dependent. Be precise. You could invoke rank to answer this question: rank < full rank therefore it doesn't span $\mathbb{R}^3$. This argument follows from linear dependence

The vectors span the subspace $\{ \mu u_1 + \lambda u_4 \}$, i.e. the points

$$r = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \mu + \lambda \\ \mu \\ \mu + \lambda \end{pmatrix} \Rightarrow \mu = y, \quad x = \lambda + y = z$$

So any ~~points~~ ^{vectors} not in the plane $x=z$ aren't in the span of $u_1, \dots, u_5$.

Correct

6c) Recall that $\det(A) \neq 0$ only if the rows ~~and~~ of A are linearly independent. But any 3×3 matrix constructed from u_i would be reduced-rank, i.e. have dependent rows. So their determinant is all zero: they map points in $\mathbb{R}^3$ to ~~the~~ a plane, scaling volumes by a factor zero.

Yes, logic correct. But you never explain why it would have dependent rows. Because always row 1 = row 3

6a) They'll both be true, since tr and det both true.

$$b) \det(A) = 6; \quad \text{tr}(A) = 5.$$

$$\lambda_1'' \lambda_2$$

$$\lambda_1'' + \lambda_2$$

$$\text{So } \lambda_1 = 2, \lambda_2 = 3.$$

c) The eigenvalues λ solve $A \cdot x = \lambda x$, with char. polynomial $(A - \lambda I) \cdot x = 0$. We have nontrivial solutions to this homog.

$$\text{eq'n only if } \det(A - \lambda I) = 0 \Rightarrow (-1 - \lambda)(6 - \lambda) + 12 = 0.$$

$$\text{So } \lambda_1 = 2, \lambda_2 = 3, \text{ which is solved by } \lambda = 2 \text{ or } \lambda = 3.$$

$$\therefore \begin{pmatrix} -1 & 2 \\ -6 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow -3x_1 + 2x_2 = 0, -6x_1 + 4x_2 = 0.$$

So $(2, 3)^T$ is an eigenvector

$$\begin{pmatrix} -1 & 2 \\ -6 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 3 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow -4x_1 + 2x_2 = 0$$

So $(1, 2)^T$ is an eigenvector

$$\text{Let } S = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}; \quad S^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

$$\text{Then } S^{-1}A = \begin{pmatrix} 4 & -2 \\ -9 & 6 \end{pmatrix}; \quad S^{-1}AS = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = D \text{ as req'd.}$$

$$\text{ad) } A = SDS^{-1}. \quad \text{So } A^{10} = SD^{10}S^{-1} \text{ since } S^{-1} \cdot S = I \text{ cancel.}$$

$$\text{But } D^{10} = \begin{pmatrix} 2^{10} & 0 \\ 0 & 3^{10} \end{pmatrix}. \quad \text{So } A^{10} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2^{10} & 0 \\ 0 & 3^{10} \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

$$= \begin{pmatrix} 2^{11} & 2 \times 3^{10} \\ 3 \times 2^{10} & 2 \times 3^{10} \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 2^{12} - 3^{11} & 2 \times 3^{10} - 2^{11} \\ 3 \times 2^{11} - 2 \times 3^{11} & 3 \times 2^{10} + 4 \times 3^{10} \end{pmatrix}$$